Digital Health Hackathon 2026 | Technology One-Pager

Digitally-Enabled Adherence Platform (DEAP)

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The Problem

Not taking medication as prescribed is one of the biggest hidden drivers of poor health outcomes, and it hits young transplant patients especially hard. For an adolescent with a kidney transplant, missed immunosuppressant doses can lead to organ rejection, hospitalization, and death. At Cincinnati Children's, nearly half of kidney transplant patients and caregivers report at least one barrier to taking their medication, and those who do are nearly three times more likely to experience acute rejection. For a transplant family, a missed dose is not a minor lapse; it can mean rejection of a donated organ or a life-threatening event. Yet barriers are rarely assessed in a structured way, and there is no standard method to deliver proven strategies to help patients overcome them between visits.

Current Stage

DEAP is built on the team's clinic-based expertise, which was shown to reduce kidney transplant rejection episodes while actually saving money. DEAP is software-only, with no expensive proprietary hardware, connecting instead to the communication tools patients already use, and it personalizes the timing and content of messages to each patient's barriers, cognitive load, language, and schedule. The clinical foundation is proven; the opportunity now is the experience that carries it to families at scale.

Jumping Off Points for Hackathon Teams

The opportunity is the patient-and-family-facing experience layer on top of the evidence-based clinical content. Possible starting points:

- A barrier-assessment and onboarding experience that helps a patient or caregiver quickly identify what is getting in the way
- A personalization layer that adapts message timing and content to a patient's situation and preferences
- Engagement and re-engagement features: reminders, nudges, streaks, progress visualization
- A patient-facing dashboard that makes adherence progress easy to see
- A clinician-facing view showing which patients need attention and which interventions work
- Modern visual design, accessibility, and mobile experience

Things to Consider, or Not!

- The evidence-based clinical content and adherence strategies stay as is. The opportunity is the experience layer that delivers them
- A self-contained prototype with sample data is a strong one-day target; no integration with live patient data is needed
- Choose the surface that fits your team's strengths. You do not need to build all of it

Why Choose This Project

You will work on a platform built on a clinic-proven program that reduced organ rejection while saving money, and build the experience that helps it reach families every day. The problem is enormous (about half of patients with chronic conditions do not take medications as prescribed, at a cost of roughly $528 billion and 125,000 lives a year), and the surface is squarely consumer-facing: messaging, personalization, dashboards, and engagement. The team is engaged and available the morning of the event to brief teams and answer questions.